Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/17/2019
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Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	11/16/2015	**End Date:**	03/31/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2021 due to change in Principal Investigator, per PI and U. Israelsson/JPL (Ed., 9/13/21) NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2022 per U. Israelsson/JPL (Ed., 9/13/18) NOTE: End date changed to 11/30/2018 per M. Lee/NASA HQ (Ed., 2/16/16)		
Key Personnel Changes/Previous PI:	EDITOR'S NOTE (December 2015): PI Chris Oates took over the project from Thomas O'Brian on 11/16/2015.		
COI Name (Institution):	Jefferts, Steven Ph.D. (National Institute of Standards & Technology)		
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

EDITOR'S NOTE (December 2015): Principal Investigator (PI) Chris Oates took over the project from Thomas O'Brian on 11/16/2015.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10^-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10^-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10^-17 or better.

• Test of gravitation redshift at 2 x 10^-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10^-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10^-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10^-16 or better.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 9 x 10^-19.

• Two ytterbium lattice optical clocks with uncertainties better than 2 x 10^-18.

• Strontium lattice optical clocks with uncertainties of 2 x 10^-18 (at JILA). A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.)

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons with demonstrated instability below 10^-16.

• A world-class time scale comprising eight hydrogen masers and a collection of Cs clocks.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Develop and install precision time links between the MWL hardware site and NIST clocks. $80k for partial support of staff time.

Gap 12 Mo. (2018-9): Assuming successful launch in 2020, will be prepared to participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. Develop a frequency-based time scale for ACES frequency comparisons. $80k for partial support of staff time.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The major benefits and impacts of this research program can be grouped into three main areas:
1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements.

2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.

3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such as search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (November 2018 through November 2019) work has focused on two main tasks:
1. Development of a frequency-based timescale for the Microwave Link Ground Terminal (MWL GT) and the ACES mission

2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics

Progress on each task is described below.

1. Development of a frequency-based timescale for the Microwave Link Ground Terminal (MWL GT) and the ACES mission

A critical aspect of the ACES project will be the comparison between the ACES clocks and the best ground clocks on Earth via the Ground Terminal links. At NIST we have a collection of atomic clocks, both microwave and optical (described in Part 2). In particular, NIST's Time Scale, UTC (NIST), which is based on a collection of hydrogen masers and commercial cesium clocks (calibrated by Primary Standards), will be a focal point of these comparisons. Previously we had constructed a reliable, high-performing hardware connection between incoming ACES signals (via the Microwave Link Ground Terminal) and UTC (NIST). The focus this year was to upgrade the Time Scale itself. To this end, we have begun construction of a frequency-based time scale for application to the ACES mission.

The current version of the NIST timescale is designed to be "on time" with respect to international atomic time (TAI). This implies that the timescale algorithm is allowed to take frequency steps and jumps with the requirement that the time steps are allowed are small, and the aggregate of all the time steps are not to exceed more than ~50 ps a day. However, for ACES, it would be better to have a time scale based on a stable frequency source. Indeed, having a continuous frequency output based on the ensemble of clocks would enable us to construct a stable frequency source with almost 100% duty cycle which would enable us to compare the two non-collocated clocks very accurately. This would enable us to test (and constrain) frequency transfer noise and more readily participate in frequency-based tests of fundamental physics.

For the new time scale, the four-parameter steering algorithm we are implementing keeps the number of time steps to a minimum and the time continuous. However, the difference between the local clock and TAI may not be as stable (or at least, we do not yet have the data to show it is) as the current NIST timescale. Still, maintaining smooth frequency is the goal, and critically, this new system can be constructed using the existing infrastructure at NIST.

A final important feature in our implementation includes data integrity checks for distinguishing clock noise from measurement noise by using clock difference data to detect cycle slip and outliers and hopefully to partially repair the data using predictions from Kalman-like filters. This new algorithm should generate a frequency scale that will be robust and stable over the duration comparable to each pass of the International Space Station (ISS) over the site location. As a result, we should be able to reduce the uncertainties in the estimate for clock frequencies for time and frequency transfer using the MWL.

The development of this time scale requires new algorithms for: (1) changing clock weights, (2) steering to externally determined primary or secondary frequency standards--either time or frequency, (3) losing or adding a clock, (4) determining optimum weights when data is correlated, (5) repairing data files with missing data or cycle slips, and (6) constructing Kalman-like filters for following the frequency and time of a clock, with frequency measurements only available. We have implemented first versions of these algorithms and spent a good portion of the past year testing them. This is a time-consuming process because tests need to be conducted over many months to get meaningful results. Still, we have made good progress and have tested both frequency steers and data repairs with no roadblocks yet identified. We anticipate having the new time scale operational well in advance of the ACES launch date.

2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics.

NIST continues to improve the performance of its multiple optical and microwave frequency standards, which will be crucial to international time and frequency comparisons and ACES-mediated fundamental research. The past project year included advances in several of NIST's frequency standards:

• The NIST Al+ optical clock has reduced its major sources of uncertainty, and in 2019 the team reported in Physical Review Letters a total uncertainty budget for the Al+ clock of 9 x 10^-19, the first clock worldwide to report an uncertainty of less than 1 x 10^-18.

• The NIST ytterbium lattice optical clocks have a final uncertainty budget with a fractional frequency uncertainty to below 1.4 x 10^-18, a level they now reachable in a few hours of averaging time. A month-long comparison between the two Yb clocks demonstrated agreement at the 7 x 10^-19 level, more than a factor of five beyond current state-of-the-art reproducibility. Taking advantage of the Yb clock's high stability, the comparison also reached a instability of 3 x 10^-19. This remarkable stability dramatically advances the goals of ACES time and frequency transfer given the relatively short visibility of the ISS to any MWL GT.

• Sr lattice optical clocks at JILA have demonstrated fractional frequency uncertainty of 2.0 x 10^-18. The JILA Sr team recently reported a record instability for their Sr clock of 4.8 x 10^-17 at one second averaging time. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and inter-comparisons.

• Recent comparisons between the Yb lattice clocks and the NIST Time Scale have enabled NIST to report the most accurate absolute frequency measurement to date, with a fractional uncertainty of 2.1 x 10^-16. In conjunction with other measurements worldwide, the loop of Cs-Yb-Sr-Yb measurements is now consistent at the limit of the current realization of the SI (International System of Units) second, which represents an important milestone on the BIPM (Bureau International des Poids et Mesures) Roadmap for the Redefinition of the SI Second. The Yb optical system is now periodically contributing to international time.

• The NIST Calcium Thermal Beam Clock reported in Physical Review Letters a fractional frequency instability below 1 x 10^-16, more than a factor of 40 lower than any other thermal beam standard worldwide. This compact clock system could serve as a prototype for a first commercial optical clock.

• The NIST clock teams have completed a series of frequency ratio measurements between the optical frequency standards (Yb, Sr, Al+) with uncertainties well below 1 x 10^-17, which have also enabled tests of drifts of fundamental constants.

In summary, at the time of this report, NIST (including NIST/JILA) currently operates two of the best microwave primary frequency standards (Cs fountains) and four leading optical standards (Yb and Sr neutral atom lattice clocks, Al ion clocks, and a Ca thermal beam clock). As highlighted above, NIST is also developing new microwave time scales that will microwave time comparison and time transfer. Other laboratories across the world are achieving high performance as well, and the NIST optical frequency standards are likely to continue to improve. This dramatic improvement of optical frequency standard performance strongly supports ACES goals of improved international time and frequency transfer and the ability to improve fundamental tests of physics.

Bibliography Type:	Description: (Last Updated: 04/05/2023)
Articles in Peer-reviewed Journals	Rovera GD, Siccardi M, Romisch S, Abgrall M. "Time delay measurements: estimation of the error budget." Metrologia. 2019 May:56(3):035004. https://doi.org/10.1088/1681-7575/ab14bb , May-2019
Articles in Peer-reviewed Journals	McGrew WF, Zhang X, Fasano RJ, Schaffer SA, Beloy K, Nicolodi D, Brown RC, Hinkley N, Milani G, Schioppo M, Yoon TH, Ludlow AD. "Atomic clock performance enabling geodesy below the centimeter level." Nature. 2018 Dec;564(7734):87-90. https://doi.org/10.1038/s41586-018-0738-2 ; PubMed PMID: 30487601 , Dec-2018
Articles in Peer-reviewed Journals	Yao J, Sherman J, Fortier T, Leopardi H, Parker T, Levine J, Savory J, Romisch S, McGrew W, Zhang X, Nicolodi D, Fasano R, Schaffer S, Beloy K, Ludlow A. "Progress on optical-clock-based time scale at NIST: simulations and preliminary data analysis." Navigation. 2018 Winter:65(4):601-8. https://doi.org/10.1002/navi.248 , Dec-2018
Articles in Peer-reviewed Journals	McGrew WF, Zhang X, Leopardi H, Fasano RJ, Nicolodi D, Beloy K, Yao J, Sherman JA, Schaffer SA, Savory J, Romisch S, Oates CW, Parker TE, Fortier TM, Ludlow AD. "Towards the optical second: verifying optical clocks at the SI limit." Optica. 2019 Apr;6(4):448-54. https://doi.org/10.1364/OPTICA.6.000448 , Apr-2019
Articles in Peer-reviewed Journals	Brewer SM, Chen JS, Hankin AM, Clements ER, Chou CW, Wineland DJ, Hume DB, Leibrandt DR. "(27)Al+ Quantum-logic clock with a systematic uncertainty below 10(-18)." Physical Review Letters. 2019 Jul 19;123(3):033201. https://doi.org/10.1103/PhysRevLett.123.033201 ; PubMed PMID: 31386450 , Jul-2019
Articles in Peer-reviewed Journals	Brewer SM, Chen JS, Beloy K, Hankin AM, Clements ER, Chou CW, McGrew WF, Zhang X, Fasano FJ, Nicolodi D, Leopardi H, Fortier TM, Diddams SA, Ludlow AD, Wineland DJ, Leibrandt DR, Hume DB. "Measurements of (27)Al+ and (25)Mg+ magnetic constants for improved ion clock accuracy." Physical Review A. 2019 Jul;100:013409. https://doi.org/10.1103/PhysRevA.100.013409 , Jul-2019
Articles in Peer-reviewed Journals	Olson J, Fox RW, Fortier TM, Sheerin TF, Frown RC, Leopardi H, Stoner RE, Oates CW, Ludlow AD. "Ramsey-Borde matter wave interferometry for laser frequency stabilization at 10(-16) frequency instability and below." Physical Review Letters. 2019 Aug 16;123(7):073202. https://doi.org/10.1103/PhysRevLett.123.073202 ; PubMed PMID: 31491125 , Aug-2019
Articles in Peer-reviewed Journals	Milner WR, Robinson JM, Kennedy CJ, Bothwell T, Kedar D, Matei DG, Legero T, Sterr U, Riehle F, Leopari H, Fortier TM, Sherman JA, Levine J, Yao J, Ye J, Oelker E. "Demonstration of a time scale based on a stable optical carrier." Physical Review Letters. 2019 Oct 25;123(17):173201. https://doi.org/10.1103/PhysRevLett.123.173201 [Note: previously reported in September 2019 as "accepted. https://www.researchgate.net/publication/334316842"] , Oct-2019